

Project 1: Home Network to ISP to Internet

NOC Portfolio Project Documentation

Author: Hassan Khan

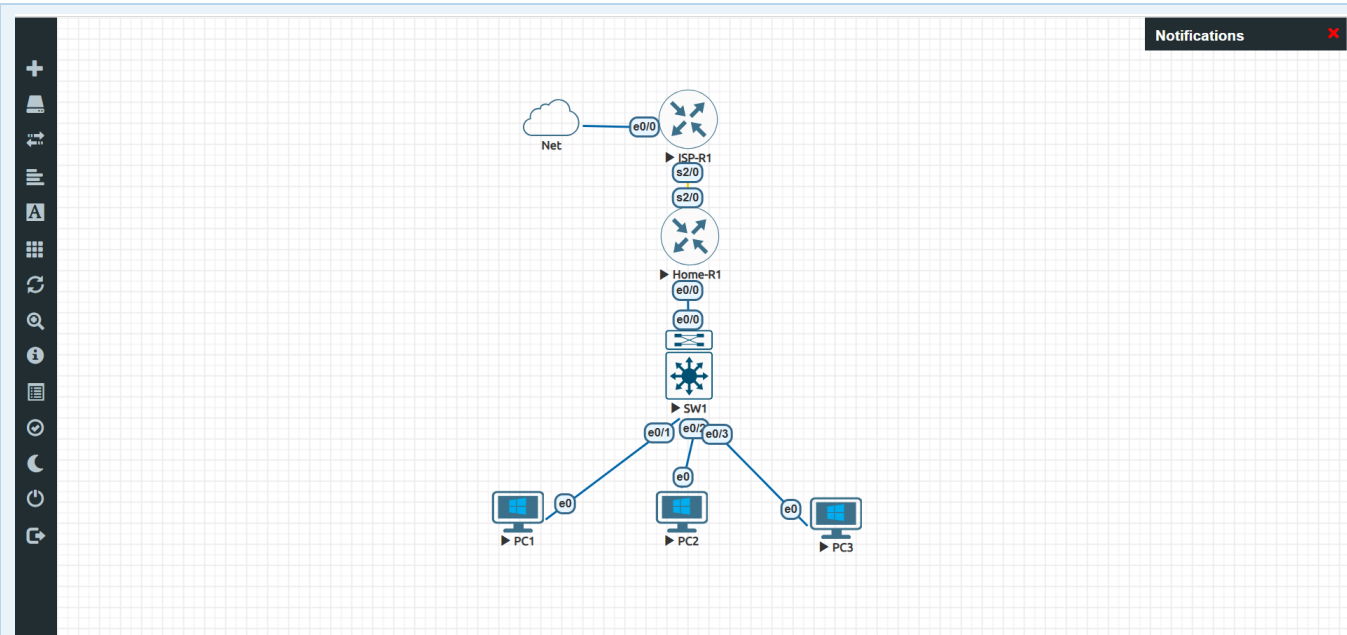
Date: 07/26/2026

Environment: EVE-NG Community Edition

1. Project Objective

Design, build, and test a simulated network that models a real home-to-internet path: home LAN → home router (NAT) → ISP → internet with every device's role and every OSI layer interaction documented and proven with live testing, not just described.

2. Topology



SCREENSHOT 1 insert here: full EVE-NG canvas view of the topology (all 6 nodes, all cables, all interface labels visible).

Devices

Device	Role	Platform
Cloud0	Real internet uplink (NAT-type EVE-NG cloud)	Built-in EVE-NG object
ISP-R1	Simulated ISP edge router, does the second NAT hop out to the real internet	Cisco IOL (router)
Home-R1	Home router NAT + DHCP for the LAN	Cisco IOL (router)
SW1	Layer 2 access switch	Cisco IOL (switch)
PC1	LAN client	Windows XP
PC2	LAN client	Windows XP
PC3	LAN client	Windows XP

Cabling

From	Port	To	Port
Cloud0 (NAT)	—	ISP-R1	e0/0
ISP-R1	s2/0	Home-R1	s2/0
Home-R1	e0/0	SW1	e0/0
SW1	e0/1	PC1	Local Area Connection
SW1	e0/2	PC2	Local Area Connection
SW1	e0/3	PC3	Local Area Connection

The ISP-R1 ↔ Home-R1 link uses serial interfaces (leased-line simulation), not Ethernet everything else is switched/routed Ethernet.

3. IP Addressing

Segment	Network	Device	Address
ISP uplink (real internet)	DHCP-assigned by NAT cloud	ISP-R1 (e0/0)	dynamic
ISP ↔ Home WAN link	203.0.113.0/30	ISP-R1 (s2/0)	203.0.113.1
ISP ↔ Home WAN link	203.0.113.0/30	Home-R1 (s2/0)	203.0.113.2
Home LAN	192.168.1.0/24	Home-R1 (e0/0)	192.168.1.1 (gateway)
Home LAN	192.168.1.0/24	PC1	DHCP-assigned
Home LAN	192.168.1.0/24	PC2	DHCP-assigned
Home LAN	192.168.1.0/24	PC3	DHCP-assigned

DNS server handed out to all three PCs via DHCP: 8.8.8.8

4. Device Configuration Summary

- Home-R1 NAT (inside/outside) + DHCP pool for 192.168.1.0/24, default route to ISP-R1.
- ISP-R1 second NAT hop (inside/outside) translating 203.0.113.0/30 out through the DHCP-assigned uplink to Cloud0, giving the whole LAN real, working internet access. This is a double-NAT design, matching how a real home router behind a real ISP's own NAT works.
- SW1 plain Layer 2 access switch, ports e0/1–e0/3 in VLAN 1.

Full CLI Configuration

Home-R1

```
enable
configure terminal
hostname Home-R1
interface Serial2/0
description WAN-to-ISP
ip address 203.0.113.2 255.255.255.252
ip nat outside
no shutdown
interface Ethernet0/0
description LAN-to-Switch
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
!
```

```
ip nat inside source list 1 interface Serial2/0 overload
access-list 1 permit 192.168.1.0 0.0.0.255
!
ip dhcp pool LAN-POOL
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1
end
write memory
```

ISP-R1

```
enable
configure terminal
hostname ISP-R1
interface Ethernet0/0
description Uplink-to-NAT-cloud
ip address dhcp
ip nat outside
no shutdown
interface Serial2/0
description Link-to-Home-R1
ip address 203.0.113.1 255.255.255.252
ip nat inside
no shutdown
!
ip nat inside source list 1 interface Ethernet0/0 overload
access-list 1 permit 203.0.113.0 0.0.0.3
end
write memory
```

SW1

```
enable
configure terminal
hostname SW1
interface range Ethernet0/1-3
switchport mode access
switchport access vlan 1
no shutdown
end
write memory
```

5. OSI Layer Mapping

This maps one real, tested event a PC resolving and reaching example.com to every layer it touches.

Layer	Name	What happens in this lab	Evidence
1	Physical	Signal travels PC → SW1 → Home-R1 over Ethernet, then Home-R1 → ISP-R1 over the serial WAN link, then ISP-R1 → real internet through the NAT cloud	Link status, show interfaces serial 2/0
2	Data Link	Ethernet framing with MAC addresses on the LAN segments; HDLC framing on the serial WAN link, re-encapsulated at every hop	show interfaces, ARP table
3	Network	Packet is routed and NAT'd twice: 192.168.1.x → 203.0.113.2 at Home-R1 → a real address at ISP-R1 → out to the internet	show ip route, show ip nat translations (both routers)
4	Transport	DNS query over UDP/53 to 8.8.8.8; ICMP echo request/reply for the ping tests	Packet capture, nslookup output
5, 6	Session / Presentation	Not separately visible with plain DNS/ICMP in this test	n/a
7	Application	example.com resolves to a real IP via DNS; the client then addresses its traffic to that IP by name	nslookup response

6. Connectivity Test Results

All tests below were run from PC1, PC2, and PC3 and succeeded on all three.

Test 1 Gateway reachability (Layer 1 to 3, LAN only)

```
ping 192.168.1.1
```

Proves: physical link, switching, and the default gateway on Home-R1 all work.

```

C:\WINDOWS\system32\cmd.exe

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>S
'S' is not recognized as an internal or external command,
operable program or batch file.

C:\Documents and Settings\user>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=2ms TTL=255
Reply from 192.168.1.1: bytes=32 time=3ms TTL=255
Reply from 192.168.1.1: bytes=32 time=2ms TTL=255
Reply from 192.168.1.1: bytes=32 time=2ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

```

SCREENSHOT 2 insert here: successful ping 192.168.1.1 output from PC1 (or all three, if you want three separate shots).

Test 2 WAN hop reachability (Layer 3, across NAT)

```
ping 203.0.113.1
```

Proves: Home-R1's routing and outbound NAT correctly forward traffic across the serial WAN link to ISP-R1.

```
C:\WINDOWS\system32\cmd.exe
Reply from 192.168.1.1: bytes=32 time=2ms TTL=255
Reply from 192.168.1.1: bytes=32 time=3ms TTL=255
Reply from 192.168.1.1: bytes=32 time=2ms TTL=255
Reply from 192.168.1.1: bytes=32 time=2ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 203.0.113.1

Pinging 203.0.113.1 with 32 bytes of data:

Reply from 203.0.113.1: bytes=32 time=35ms TTL=254
Reply from 203.0.113.1: bytes=32 time=31ms TTL=254
Reply from 203.0.113.1: bytes=32 time=31ms TTL=254
Reply from 203.0.113.1: bytes=32 time=31ms TTL=254

Ping statistics for 203.0.113.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 31ms, Maximum = 35ms, Average = 32ms

C:\Documents and Settings\user>
```

 SCREENSHOT 3 insert here: *successful ping 203.0.113.1 output.*

Test 3 Real internet reachability by IP (Layer 3 only, no DNS)

```
ping 8.8.8.8
```

Proves: the double-NAT path (Home-R1 → ISP-R1 → Cloud0 → real internet) works end to end at the IP layer, independent of DNS.

```
C:\WINDOWS\system32\cmd.exe
Reply from 203.0.113.1: bytes=32 time=35ms TTL=254
Reply from 203.0.113.1: bytes=32 time=31ms TTL=254
Reply from 203.0.113.1: bytes=32 time=31ms TTL=254
Reply from 203.0.113.1: bytes=32 time=31ms TTL=254

Ping statistics for 203.0.113.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 31ms, Maximum = 35ms, Average = 32ms

C:\Documents and Settings\user>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=126ms TTL=126
Reply from 8.8.8.8: bytes=32 time=142ms TTL=126
Reply from 8.8.8.8: bytes=32 time=1302ms TTL=126
Reply from 8.8.8.8: bytes=32 time=164ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 126ms, Maximum = 1302ms, Average = 433ms

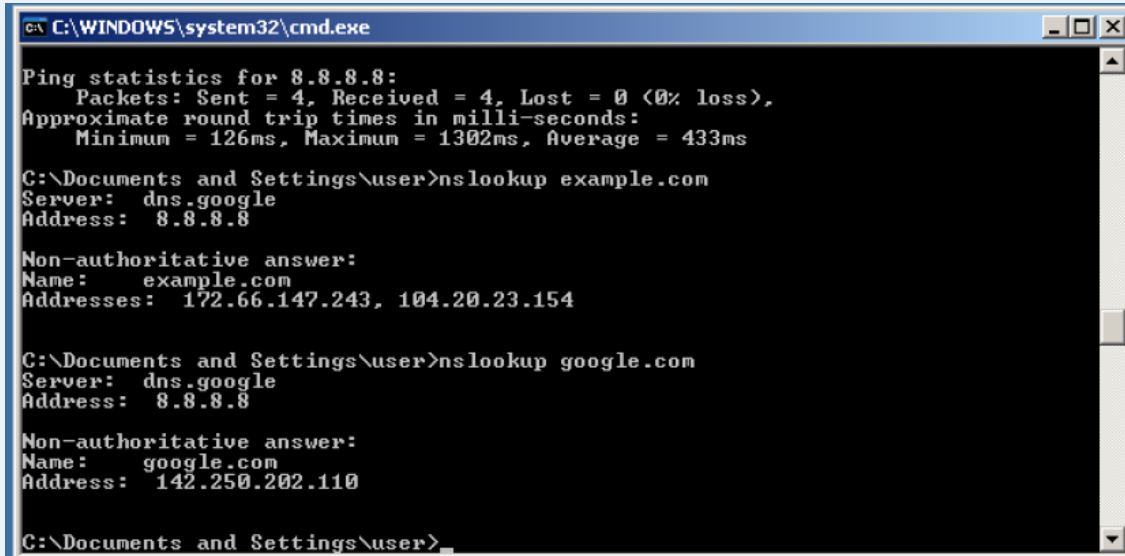
C:\Documents and Settings\user>
```

 SCREENSHOT 4 insert here: *successful ping 8.8.8.8 output.*

Test 4 Real DNS resolution (Layer 7)

```
nslookup google.com
```

Proves: the PC's resolver, pointed at 8.8.8.8 via DHCP, can reach a real public DNS server and get back a real answer.



```
C:\WINDOWS\system32\cmd.exe

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 126ms, Maximum = 1302ms, Average = 433ms

C:\Documents and Settings\user>nslookup example.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     example.com
Addresses: 172.66.147.243, 104.20.23.154

C:\Documents and Settings\user>nslookup google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Address:  142.250.202.110

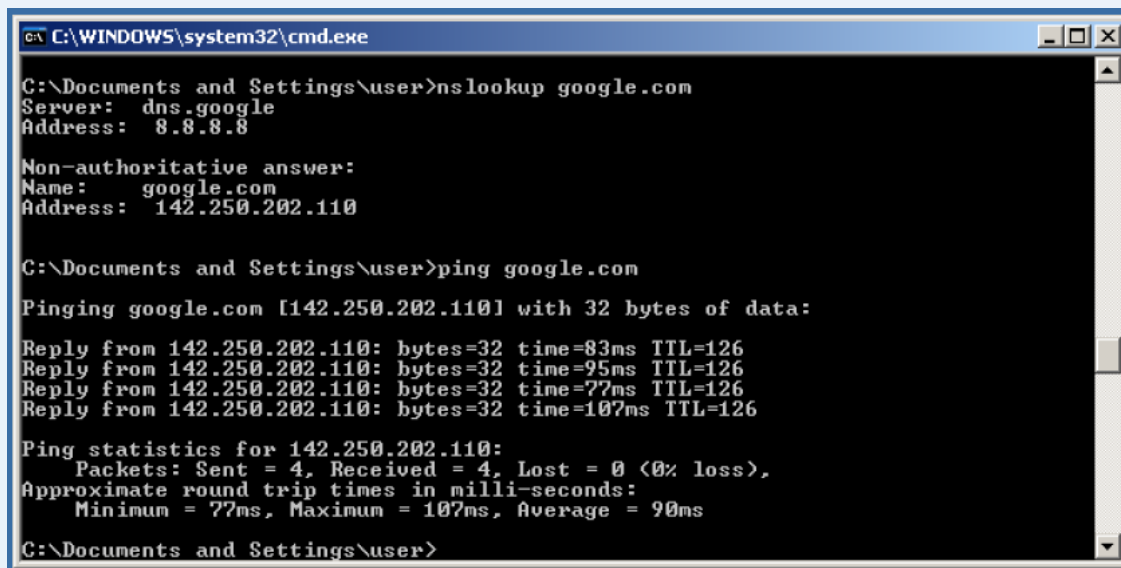
C:\Documents and Settings\user>
```

 **SCREENSHOT 5 insert here:** *nslookup example.com output, showing the response coming from server 8.8.8.8 with a resolved IP address.*

Test 5 Real internet reachability by name (Layer 3 + Layer 7 together)

ping example.com

Proves: DNS resolution and routing work together — the name resolves, and the resolved address is actually reachable across the full path.



```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>nslookup google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Address:  142.250.202.110

C:\Documents and Settings\user>ping google.com

Pinging google.com [142.250.202.110] with 32 bytes of data:

Reply from 142.250.202.110: bytes=32 time=83ms TTL=126
Reply from 142.250.202.110: bytes=32 time=95ms TTL=126
Reply from 142.250.202.110: bytes=32 time=77ms TTL=126
Reply from 142.250.202.110: bytes=32 time=107ms TTL=126

Ping statistics for 142.250.202.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 77ms, Maximum = 107ms, Average = 90ms

C:\Documents and Settings\user>
```

 **SCREENSHOT 6 insert here:** *successful ping google.com output, showing the resolved IP address responding.*

Supporting evidence (routers)

```
ISP-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override

Gateway of last resort is 192.168.149.2 to network 0.0.0.0

S*    0.0.0.0/0 [254/0] via 192.168.149.2
      192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.0.2.0/24 is directly connected, Ethernet0/1
L      192.0.2.1/32 is directly connected, Ethernet0/1
      192.168.149.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.149.0/24 is directly connected, Ethernet0/0
L      192.168.149.156/32 is directly connected, Ethernet0/0
      203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C      203.0.113.0/30 is directly connected, Serial2/0
L      203.0.113.1/32 is directly connected, Serial2/0
ISP-R1#
```

 **SCREENSHOT 7** insert here: *show ip route on ISP-R1*

```
Home-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 203.0.113.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Ethernet0/0
L      192.168.1.1/32 is directly connected, Ethernet0/0
      203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C      203.0.113.0/30 is directly connected, Serial2/0
L      203.0.113.2/32 is directly connected, Serial2/0
..
```

 **SCREENSHOT 8** — insert here: *show ip route on HOME-R1*


```
Home-R1#sh int s2/0
Serial2/0 is up, line protocol is up
Hardware is M4T
Description: WAN-to-ISP
Internet address is 203.0.113.2/30
MTU 1500 bytes, BW 1544 Kbit/sec, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, crc 16, loopback not set
Keepalive set (10 sec)
Restart-Delay is 0 secs
Last input 00:00:01, output 00:00:01, output hang never
Last clearing of "show interface" counters 02:56:25
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
 42371 packets input, 61463039 bytes, 0 no buffer
Received 932 broadcasts (0 IP multicasts)
 0 runs, 0 giants, 0 throttles
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
38984 packets output, 1758019 bytes, 0 underruns
 0 output errors, 0 collisions, 3 interface resets
 0 unknown protocol drops
 0 output buffer failures, 0 output buffers swapped out
 3 carrier transitions      DCD=up   DSR=up   DTR=up   RTS=up   CTS=up
```

Home-R1#

```
ISP-R1#show interfaces serial 2/0
Serial2/0 is up, line protocol is up
Hardware is M4T
Description: Link-to-Home-R1
Internet address is 203.0.113.1/30
MTU 1500 bytes, BW 1544 Kbit/sec, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, crc 16, loopback not set
Keepalive set (10 sec)
Restart-Delay is 0 secs
Last input 00:00:02, output 00:00:02, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 35861
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
 38592 packets input, 1728532 bytes, 0 no buffer
Received 542 broadcasts (0 IP multicasts)
 0 runs, 0 giants, 0 throttles
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
41980 packets output, 61432086 bytes, 0 underruns
 0 output errors, 0 collisions, 1 interface resets
 1 unknown protocol drops
 0 output buffer failures, 0 output buffers swapped out
 2 carrier transitions      DCD=up   DSR=up   DTR=up   RTS=up   CTS=up
```

ISP-R1#

ISP-R1#

 **SCREENSHOT 9** — insert here: *show interfaces serial 2/0 on either router, showing line protocol up and correct clocking on the WAN link.*

7. NOC Troubleshooting Notes

Points worth knowing if any of the above had failed, and worth including in a portfolio write-up to show operational understanding, not just a working build:

- Serial link down/down → almost always a missing clock rate on the DCE end of the link.
- PC gets no IP / wrong subnet → check the DHCP pool's network statement on Home-R1 and that ip nat inside/outside are on the correct interfaces (reversing them breaks NAT silently).
- ping 192.168.1.1 works but ping 203.0.113.1 fails → routing problem between Home-R1 and ISP-R1, check show ip route on both ends and confirm the serial interfaces are actually up.
- ping 8.8.8.8 fails but everything internal works → check NAT translations on ISP-R1 specifically (the second hop), and confirm the EVE-NG host itself has real internet access to hand down through the NAT cloud.
- nslookup fails but ping 8.8.8.8 succeeds → DNS-specific issue: confirm the DHCP pool is actually handing out 8.8.8.8 and that UDP/53 isn't being blocked anywhere in the NAT/ACL chain.

8. Conclusion

This project demonstrates a complete, working simulation of a home network reaching the real internet through a simulated ISP, including:

- A realistic double-NAT design matching how real home routers and ISPs actually operate
- A serial WAN link matching real leased-line ISP handoffs
- Full OSI-layer traceability for a real DNS + connectivity event, backed by live test evidence rather than just diagrams
- Genuine DNS resolution against a real public resolver, with no fabricated or local-only servers involved

Skills demonstrated:

Cisco IOS routing and NAT configuration, DHCP server configuration, Layer 2 switching, WAN/serial link configuration and troubleshooting, OSI model mapping to real traffic, and structured technical documentation.